AI Assisted Coding

LabAssignment14.3

Name : J. Sujith

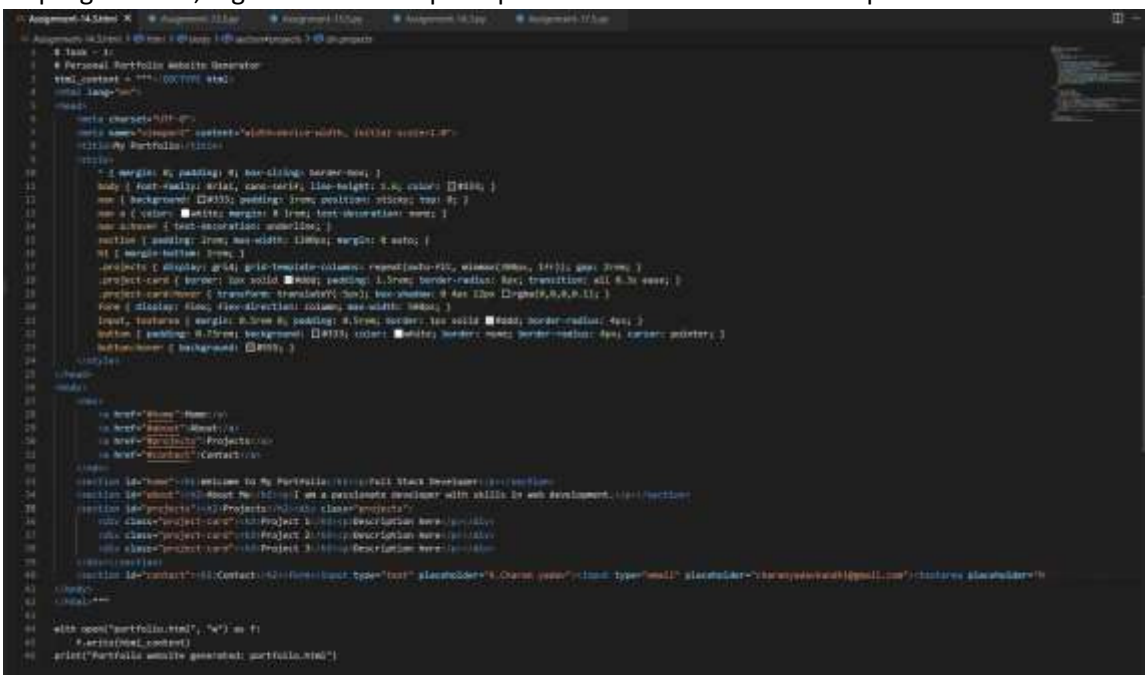
Hall Ticket no : 2303A51327

Batch No : 20

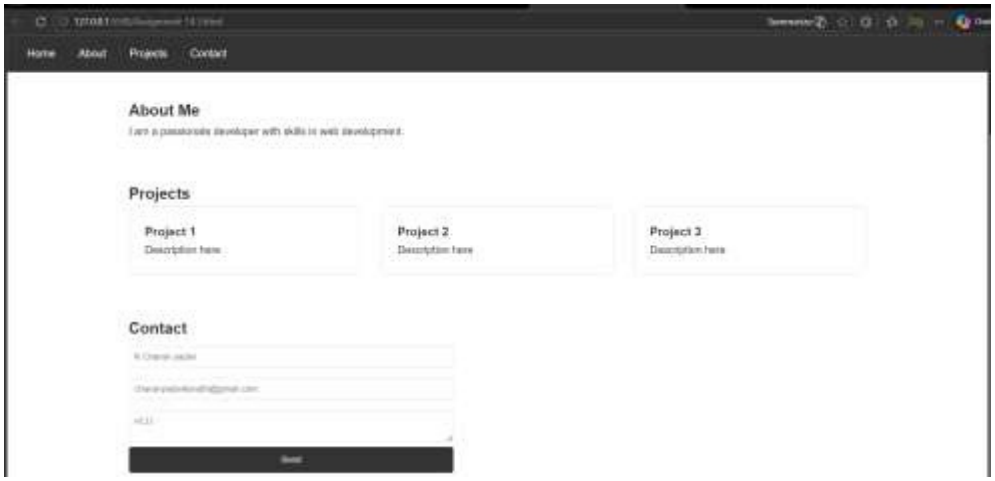
Task -1:

Prompt:

The taskistodevelop a Personal Portfolio Website Generator using Python that automatically creates an HTMLportfoliowebpage. The website includes a navigation bar, About section, Projects section, and Contactform.The HTML and CSS are generated using a Python script and saved as an HTML file. When theprogramruns, it generates a complete portfolio website that can be opened in a browser.



OUTPUT:

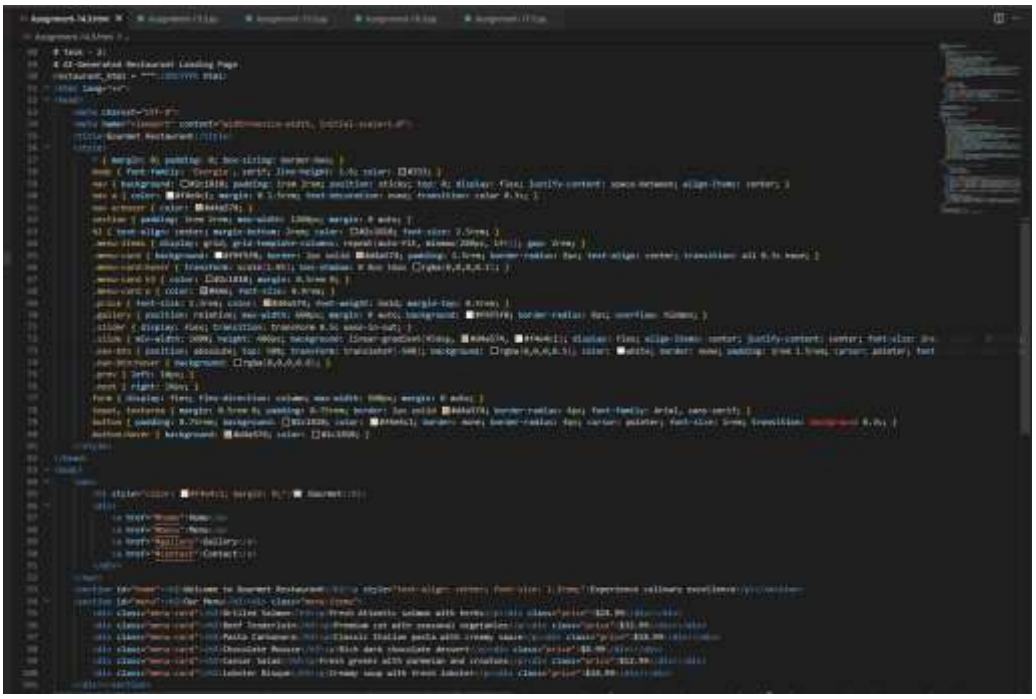


Justification: This project shows how Python can be used to automate website creation using file handling. It reduces manual work by generating HTML and CSS automatically. The project demonstrates integration of Python with web development concepts like HTML structure, CSS styling, and forms. It is useful for quickly creating portfolio websites for students and developers.

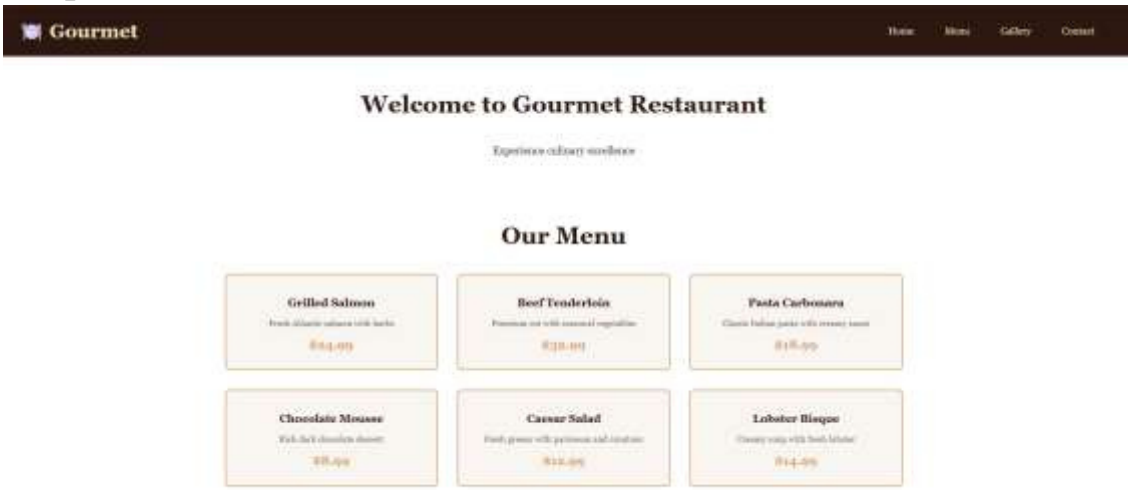
Task 2:

Prompt:

The task istodevelop an AI-Generated Restaurant Landing Page using Python that automatically creates anHTML and CSS restaurant website. The website includes a navigation bar, home section, menu sectionwith food items and prices, gallery section, and contact form. The Python script generatestheHTML content and saves it as a webpage file. The generated page should be visually styled andresponsive using CSS grid and layout design.



Output:



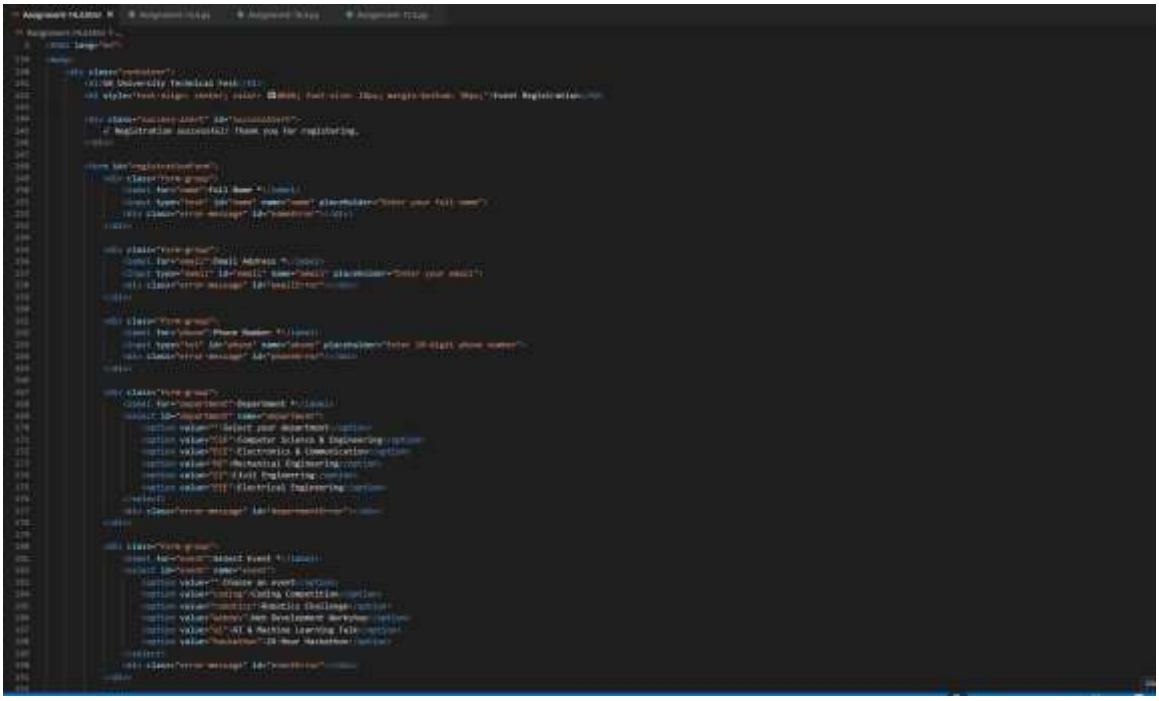
Justification:

This projectdemonstrates how Python can be used to automatically generate styled web pages using HTML and CSS. It reducesmanual webpage creation by dynamically generating the restaurant website layout and menu items. The projectshows integration of Python scripting with web development technologies. It is useful for automating landing pagecreation for restaurants or small businesses.

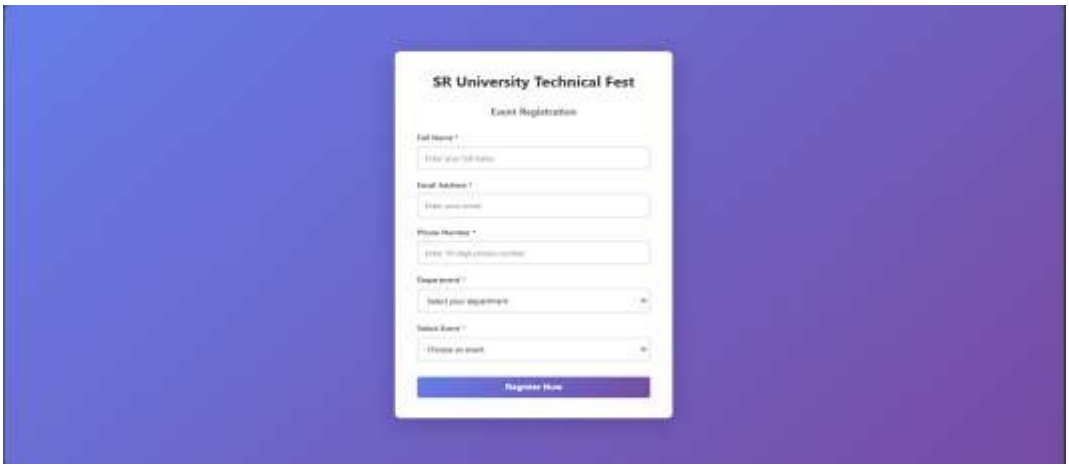
Task 3:

Prompt:

Thetask isto create an AI-Powered Event Registration Form for SR University Technical Fest using HTML, CSS, and JavaScript. The form should include fields such as Name, Email, Phone Number, Department, and Event Selection. The form must be styled using CSS for a user-friendly interface and implement JavaScript for real-time validation like email format, phone number length, and required field validation. The form should prevent submission if invalid data is entered.



Output:



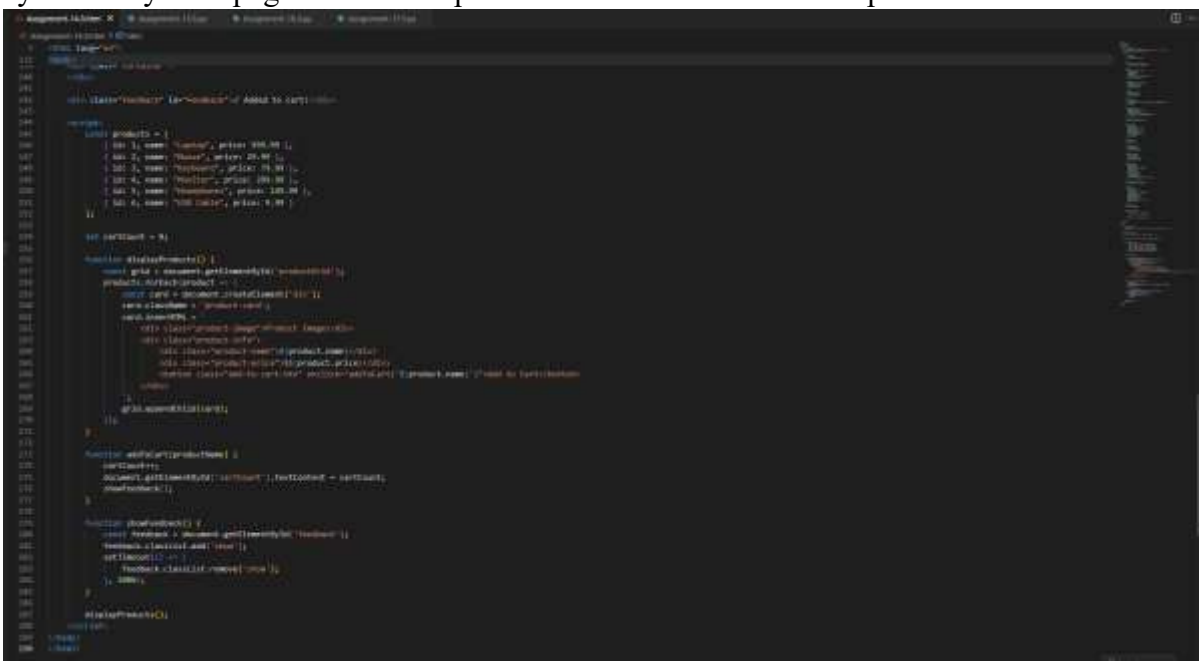
Justification:

Thisprojectdemonstrates how AI tools like Copilot can assist in generating web forms with validation and styling automatically.Itimproves user experience by providing real-time validation and preventing incorrect form submissions.Theproject integrates HTML, CSS, and JavaScript to build an interactive web form. It is useful for event registrationsandonline data collection systems.

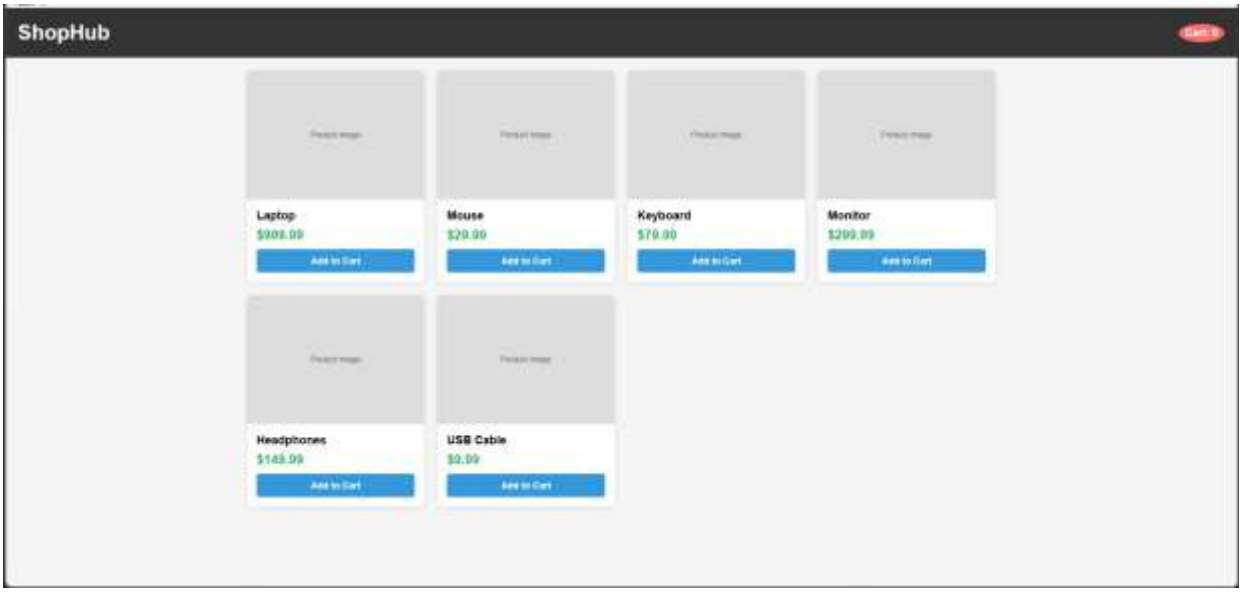
Task -4:

Prompt:

The task is todevelop an AI-Assisted E-Commerce Product Page using HTML, CSS, and JavaScript. The page should display products in a grid layout with product name, price, and an “Add to Cart” button. JavaScript should be used to implement cart functionality and update the cart item counter dynamically. The page should also provide visual feedback when a product is added to the cart.



Output:



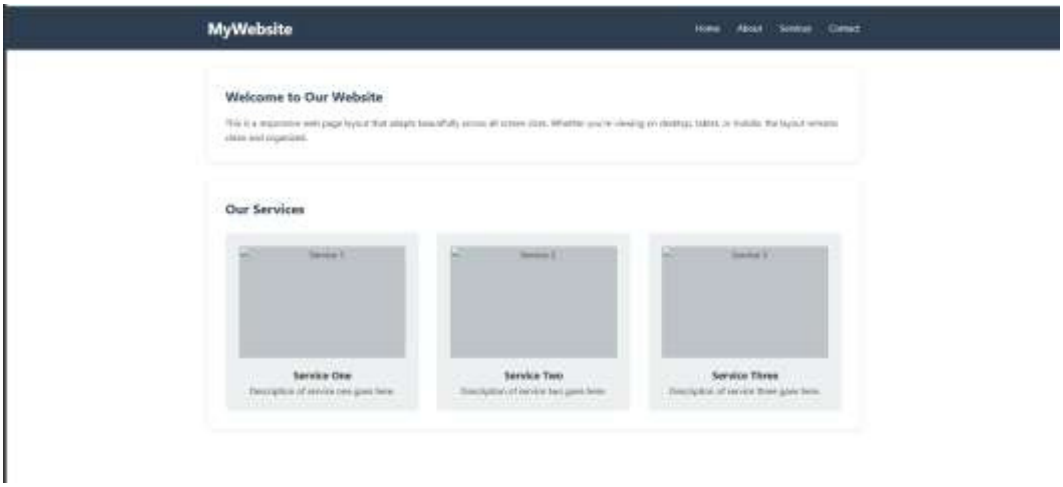
Justification:

Thisproject demonstrates how AI tools like Copilot can help generate web development code for an e-commerce product page.Itintegrates HTML for structure, CSS for layout design, and JavaScript for cart functionality and interactivity. Theproject improves user experience through dynamic cart updates and visual feedback. It is useful for understandingbasic e-commerce website functionality and front-end development concepts.

Task – 5:

Prompt: Thetask is to create a responsive web page layout using HTML and CSS that includes a header withnavigation links, a main content section, and a footer. The layout should be responsive and adapt todifferent screen sizes such as desktop, tablet, and mobile using CSS media queries. AI tools can beusedto generate responsive CSS and layout structure. The page should be clean, organized, andvisually appealing.

Output:



Justification: This project demonstrates how responsive web design can be implemented using HTML and CSS withmediaqueries. Using AI assistance helps in generating responsive layouts quickly andefficiently. The project improves understanding of layout design, navigation bars, content sections, and footers.Itis useful for building modern websites that work across multiple devices and screen sizes.